TANZANIAN

WATER WELLS PREDICTION

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OVERVIEW

Tanzania is a developing nation with a population of over 57 million people. It faces a critical water access crisis due to a significant number of non-functional water points

PROBLEM STATEMENT

Despite building water wells, Tanzania is failing to meet potable water requirements for its 57 million people .

- Manual inspections are costly and time consuming
- Many water points are non-functional
- Millions lack access to clean water

OBJECTIVES

1. Identify the key drivers of water point failures in Tanzania.
2. Build a predictive classification model to identify functional and non-functional water points.
3.Provide actionable insights to guide future water point constructions in Tanzania

DATA UNDERSTANDING

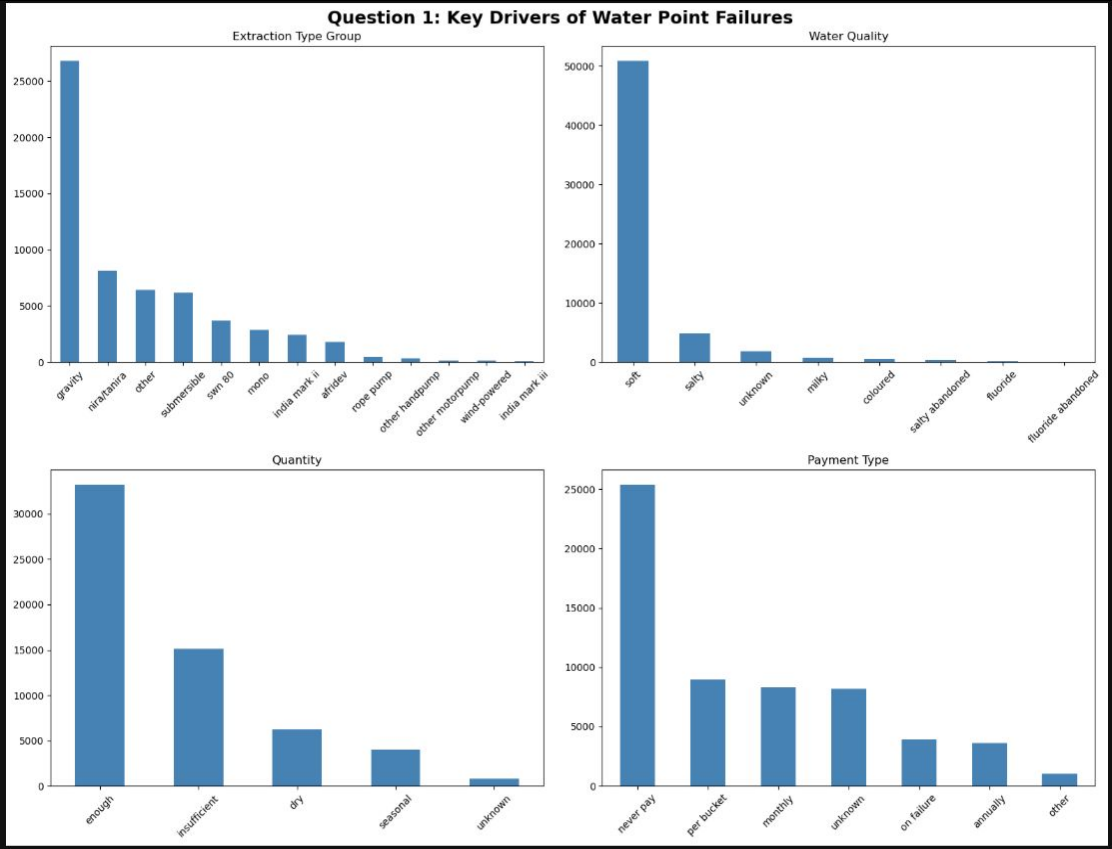
- **DATASET 59,400 Records 40 Columns**
- **TARGET VARIABLE Functional Non Functional Functional Needs Repair**

Source: Tanzanian Ministry of Water via DrivenData Competition

UNIVARIATE ANALYSIS

KEY FINDINGS:

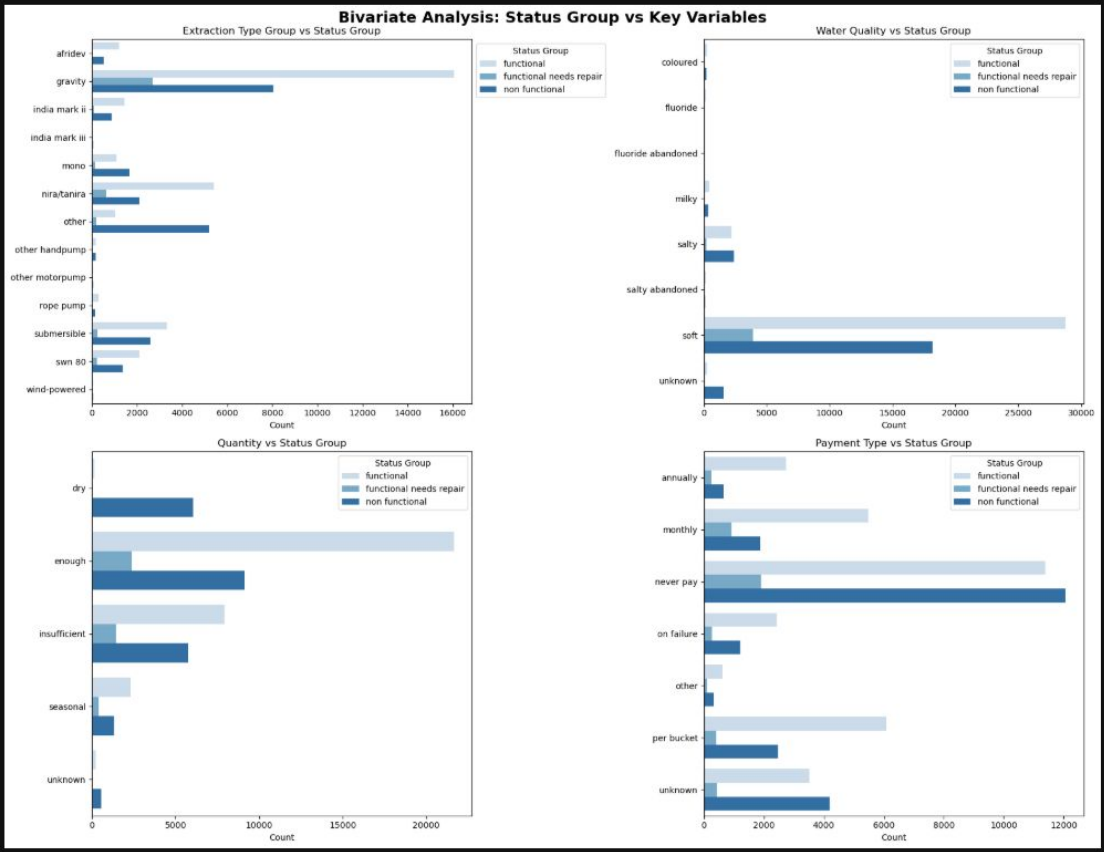
- Gravity is the most common extraction type (26,000+ wells)
- Soft water quality dominates with over 50,000 water points
- Most water points have enough water quantity
- Majority of water points are never paid for



BIVARIATE ANALYSIS

KEY FINDINGS:

- Gravity extraction type has the most functional wells
- Soft water quality is dominant in functional water points
- Wells with enough water quantity tend to be functional
- Wells that never pay tend to be non functional



Model Results

LOGISTIC REGRESSION

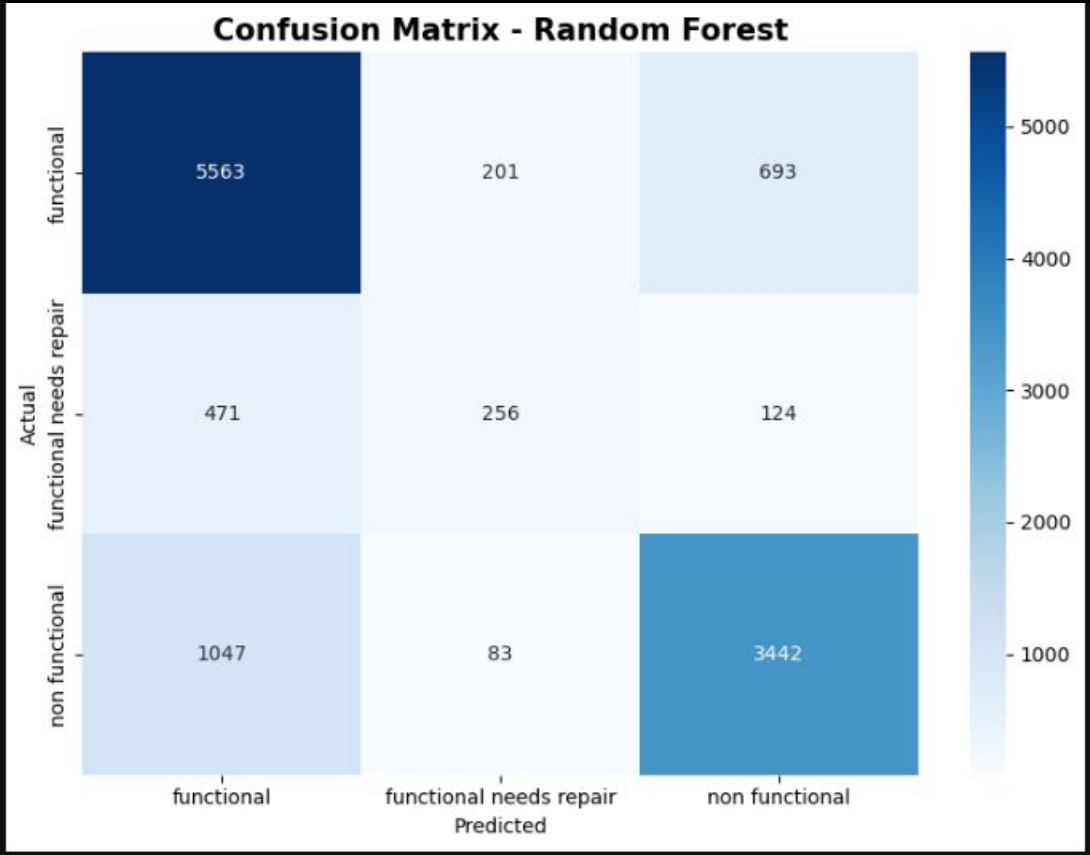
- Training Accuracy: 73%
- Testing Accuracy: 72%

DECISION TREE

- Training Accuracy: 93%
- Testing Accuracy: 75%

RANDOM FOREST

- Training Accuracy: 93%
- Testing Accuracy: 78%



RECOMMENDATIONS

1. ADDRESS CLASS IMBALANCE -Use SMOTE technique to balance functional needs repair class.
2. IMPROVE MODEL PERFORMANCE -Try advanced algorithms such as XGBoost to improve accuracy.
3. FOCUS ON HIGH FAILURE REGIONS -Prioritize regions like Dar es Salaam with fewer water points.
4. IMPROVE MANAGEMENT- Train and equip user groups with water point management skills
5. PAYMENT POLICY -Introduce affordable payment scheme to ensure maintenance of water points .
6. FOCUS ON WATER QUALITY-Invest in water treatment facilities for salty and fluoride water points

CONCLUSION

MODEL SUMMARY
- Logistic Regression: 72%
- Decision Tree: 75%
- Random Forest: 78%

TARGET ACHIEVEMENT
None of the models met our target accuracy of 80% The Random Forest model came closest at 78%"

FUTURE WORK
- Use SMOTE to address class imbalance
- Explore XGBoost for better accuracy
- Add more features to improve model

By leveraging machine learning, we can predict water point failures and ensure reliable access to clean water for the people of Tanzania

THANK YOU

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